

Viraj Ajay Mhetar

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Career Objective / Professional Summary

Undergraduate Polymer Engineering student with a Multidisciplinary Minor in AI/ML (CGPA: 8.52). Skilled in integrating material science with data-driven modeling and analytical techniques, seeking to leverage hands-on laboratory experience into a rigorous higher education program focused on data-driven sustainable polymer solutions.

Education

Institute of Chemical Technology, Mumbai, Maharashtra

Bachelor of Technology, Specialization in Polymer Engineering & Technology |

CGPA: 8.52/10 | Minor: Machine Learning & Artificial Intelligence | Honours

Pre-University Education

H.S.C. Class XII Board, Kolhapur, Maharashtra |

Percentage: 68.5%

C.B.S.E Class X Board, Kolhapur, Maharashtra |

Percentage: 96.4%

Technical Skills

Programming and Data Analysis: Python, Machine Learning, Chemical Process Simulation, Predictive/Probabilistic Modeling, Data Visualization

Laboratory / Instrumentation: FTIR, Rheometer, UTM, POM, Compound Microscopy

Core Engineering Skills: Polymer Processing (Extrusion), Protein Extraction & Isolation, Microplastic Isolation & Biomagnification Analysis, Chemical Recycling (Depolymerization)

In-Plant Training / Industrial Internship

Kothari Petrochemicals, Project Intern | May 2026 – July 2026

- Tasked with developing data-driven models for polymer grade mapping.
- Applied Artificial Intelligence and Machine Learning (AI/ML) tools using Python to analyze and map complex datasets.
- Navigated strict confidentiality parameters (NDA) by focusing on algorithmic efficiency and data

preprocessing techniques to support quality control and product categorization.

Additional Research / Work Experience

School of Nanoscience and Technology, Shivaji University, Kolhapur,
Student Researcher | May 2025 - July 2025

Detection of Microplastics in Black Soldier Fly Larvae (BSFL) | 2nd Year Summer Technosocial Community Project

- Developed experimental protocols to dissolve organic matter and isolate microplastics from complex waste substrates.
- Utilized compound microscopy and analytical techniques to analyze the impact of microplastic biomagnification.
- **Result:** Research findings published in the Bombay Technologist journal (2026).
- Mhetar, V. A., "Detection of Microplastic fibers in Black Soldier Fly Larvae Meal," Bombay Technologist (Online ISSN: 0067-9925), 2026. [[DOI Link](#)]

ICT Mumbai, Polymer Dept.,
Student Researcher | Dec 2024 - Ongoing

Bio-Based Additives for Biopolymers | Undergraduate Research Project

- Engineered protocols for the extraction and purification of proteins from ethanol industry waste feedstock to synthesize sustainable polymer additives.
- Conducted material compounding via extrusion and evaluated structural and mechanical properties utilizing FTIR, Rheometry, UTM, and POM.
- **Result:** Manuscript detailing extraction methodology and material characterization is currently in preparation.

Academic Achievements and Awards

- **1st Prize - Industrially Defined Problem Presentation**, ACP Rangotsav, 2026. Awarded for developing and presenting a viable solution for the valorization of Unsaturated Polyester Resin (UPR) effluent.
- **NPTEL Silver Medal - Data Science for Engineers**, National Programme on Technology Enhanced Learning.
- **NPTEL Silver Medal - Mechanics of Composite Materials and Structures**, National Programme on Technology Enhanced Learning.

Positions of Responsibility

Center Placement Cell (CPC) ICT Mumbai | March 2026 – Ongoing

- **Social Media Head:** Directing the digital outreach strategy and managing official platforms to bridge communication between the student body, university administration, and external industrial/academic recruiters.
- Designing and executing targeted content campaigns to promote placement drives, internships, and professional development seminars, ensuring high visibility and student engagement.

Tata Mumbai Marathon | January 2023

- **Logistics Volunteer:** Collaborated with a large-scale logistics team to coordinate the distribution of essential hydration and recovery supplies to over 55,000 participants under high-pressure conditions.

Gardens Club, Kolhapur | May 2024

- **Environmental Volunteer:** Contributed to local biodiversity and sustainability initiatives by participating in a large-scale afforestation drive, managing the planting and maintenance of indigenous tree species.

Certifications / Workshops

- **Data Science for Engineers**, National Programme on Technology Enhanced Learning (NPTEL), 2026. *Key skills: Python-based data analysis, predictive modeling, and computational problem-solving for engineering applications.*
- **Mechanics of Composite Materials and Structures**, National Programme on Technology Enhanced Learning (NPTEL), 2026. *Key skills: Advanced structural analysis, stress distribution, and mechanical behavior of complex material systems.*
- **Polymer Reaction Engineering**, National Programme on Technology Enhanced Learning (NPTEL), 2024. *Key skills: Reaction kinetics, step-growth/chain-growth mechanisms, and chemical reactor design*

Languages

English: Professional working proficiency

Hindi: Fluent

Marathi: Fluent

Spanish: Elementary proficiency (Currently learning)

Interests

3D Modeling, Competitive Chess, Swimming, Technical & Non Technical Writing

